

RIO BRANCO, Brazil

WMO: 829170

Lat: 9.869S

Lon: 67.894W

Elev: 193

StdP: 99.03

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	14.8	16.2	9.1	7.4	20.5	11.9	8.9	21.4	6.2	23.4	5.3	23.6	2.3	180	0.262

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	12.2	36.1	25.1	35.1	25.0	34.2	25.0	28.7	31.2	28.1	30.6	27.6	30.4	2.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.9	29.9	27.5	23.9	29.4	27.0	23.3	29.1	94.5	31.6	91.8	30.6	89.4	30.6	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.1	4.4	DB	12.1	37.6	2.2	1.4	10.5	38.6	9.2	39.4	7.9	40.1	6.3	41.1	(4)
(5)			WB	10.7	29.1	2.6	1.3	8.9	30.1	7.4	30.9	6.0	31.6	4.1	32.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.5	26.6	26.5	26.7	26.6	25.7	25.2	25.3	26.6	27.5	27.5	27.0	26.6	(6)	
	DBStd	2.18	1.45	1.35	1.32	1.80	2.37	2.64	2.81	2.57	2.41	1.87	1.82	1.40	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	4	0	0	0	0	1	2	2	0	0	0	0	0	(9)	
	CDD10.0	6016	513	463	517	497	487	454	474	516	525	543	512	515	(10)	
	CDD18.3	2979	255	229	259	248	229	206	218	258	275	284	262	257	(11)	
	CDH23.3	25416	1875	1662	1911	1929	1788	1705	2079	2699	2868	2652	2307	1941	(12)	
(13)	CDH26.7	10778	639	547	653	726	682	726	1002	1418	1488	1241	968	688	(13)	
(14)	Wind	WSAvg	1.5	1.6	1.5	1.4	1.3	1.3	1.4	1.6	1.7	1.7	1.7	1.7	(14)	
(15) Precipitation	PrecAvg	1804	271	252	260	170	90	43	24	45	86	146	185	226	(15)	
	PrecMax	2569	511	370	401	416	208	168	59	111	176	241	386	362	(16)	
	PrecMin	1235	120	148	125	25	17	2	1	1	24	68	84	90	(17)	
	PrecStd	300	99	57	76	80	45	37	18	29	44	51	68	71	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	33.4	33.6	34.1	34.0	34.2	35.8	38.0	37.9	37.0	35.4	34.0	(19)	
		MCWB	27.6	26.5	26.6	27.2	27.0	25.8	24.8	23.5	24.9	25.8	26.6	26.8	(20)	
	2%	DB	32.8	32.1	32.2	32.9	32.8	32.9	34.1	35.9	36.1	35.2	34.1	32.8	(21)	
		MCWB	26.7	26.1	26.1	26.8	26.3	25.3	23.9	24.3	24.9	25.7	26.4	26.3	(22)	
	5%	DB	31.6	31.2	31.3	31.9	31.8	31.9	33.0	34.8	35.0	34.0	33.0	31.8	(23)	
		MCWB	26.3	25.9	26.1	26.4	25.8	25.0	24.0	23.8	24.7	25.6	26.1	26.1	(24)	
	10%	DB	30.2	30.1	30.2	30.8	30.6	30.9	32.0	33.6	33.9	32.8	31.9	30.2	(25)	
		MCWB	25.9	25.8	26.0	26.1	25.5	24.7	23.6	23.6	24.5	25.4	25.8	25.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.2	28.5	28.6	29.4	28.7	27.7	27.2	27.4	28.1	28.4	28.8	29.0	(27)	
		MCDB	31.9	30.4	30.3	31.4	31.3	30.8	31.1	32.0	32.5	32.4	32.0	31.1	(28)	
	2%	WB	28.1	27.6	27.7	28.2	27.7	26.9	26.2	26.3	26.9	27.5	27.9	28.0	(29)	
		MCDB	30.2	29.8	29.9	30.5	30.6	30.5	30.8	31.7	31.7	31.5	30.8	30.1	(30)	
	5%	WB	27.2	26.9	27.0	27.3	26.8	26.1	25.6	25.6	26.1	26.9	27.2	27.2	(31)	
		MCDB	29.7	29.6	29.8	30.2	30.0	30.1	30.4	31.3	31.8	31.1	30.5	29.6	(32)	
	10%	WB	26.4	26.1	26.3	26.5	25.9	25.4	24.8	24.9	25.5	26.2	26.5	26.4	(33)	
		MCDB	29.1	28.9	29.1	29.5	29.2	29.5	29.7	30.7	31.3	30.4	30.0	29.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	7.7	8.0	8.6	9.1	10.4	12.3	13.2	12.2	10.4	9.2	8.2	(35)	
		MCDBR	9.4	9.1	9.3	10.0	10.5	11.7	13.5	14.5	13.5	12.3	10.9	9.8	(36)	
	5% WB	MCWBR	4.2	4.1	4.1	4.7	4.7	5.1	4.8	4.5	4.2	4.2	4.3	4.4	(37)	
		MCWBR	8.5	8.3	8.4	9.1	9.8	10.9	11.9	12.7	12.2	10.8	9.7	8.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.396	0.392	0.390	0.384	0.387	0.383	0.394	0.539	0.636	0.491	0.437	0.409	(40)	
		taud	2.497	2.512	2.530	2.514	2.487	2.505	2.439	1.993	1.760	2.183	2.343	2.448	(41)	
	Ebn at Noon		945	944	930	906	873	861	857	757	713	845	901	930	(42)	
		Edh at Noon	115	113	109	106	103	98	107	175	232	156	133	120	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.52	4.34	4.50	4.56	4.36	4.42	4.89	5.00	4.96	5.02	4.88	4.50	(44)	
(45)	RadStd		0.24	0.20	0.19	0.29	0.28	0.31	0.38	0.35	0.43	0.25	0.27	0.33	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	-1.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page